Trigonometric Approach to Forward Kinematics

Summary:

This lesson introduces forward kinematics as the process of calculating where the end effector ends up given a set of joint angles. Students review the trigonometry needed (SOH-CAH-TOA) and apply it to derive the FK equations for a simplified 2-link planar arm and then in 3d. By the end of the lesson, students can calculate the predicted end-effector position for a given set of angles and verify it against the physical arm.

Objective:

By the end of this lesson, students will be able to:
● Explain forward kinematics as calculating end-effector position from known joint angles
● Apply trigonometric principles (SOH-CAH-TOA) to a series of rotating joints
● Derive the FK equations for a simplified 2-link planar arm
● Calculate the end-effector position for a given set of joint angles

Standards:

NGSS

Resources & Materials:

Desmos Example
Slideshow

Procedures:

Day 1: 2D Forward Kinematics

1. **Opening (3 min)** — Display title slide; briefly tie back to the "how did I know which angles to use?" question from the lesson opener.
2. **Defining the arm's links (7 min)** — Introduce L1 and L2 as the link lengths (e.g., 250mm each), emphasizing these are known constants because they're fixed, measured dimensions of the physical arm.
3. **Defining the joint angles (10 min)** — Introduce θ1 and θ2, explaining that θ1 is measured relative to a vertical projection above the shoulder motors, while θ2 is

measured relative to the straight-line extension of the upper arm segment. Emphasize that although these angles are variable, they are always known because the encoder at each actuator measures its rotation — a callback to the encoders concept from Lesson 1.

4. **Setting up the problem (5 min)** — Show the arm with origin (0,0) at the shoulder joint and (X,Y) marked as the unknown position of the end effector — the value the class is trying to solve for.
5. **Breaking position into components (8 min)** — Show that the end-effector position is the sum of the horizontal and vertical displacement contributed by each joint (ΔX1, ΔY1, ΔX2), and that each displacement can be represented as a triangle.
6. **Choosing the angle for each triangle (5 min)** — Show that L1 and L2 are each the hypotenuse of their own triangle, and that to use sin and cos we need the angle between each link and vertical. For the upper arm, θ1 is already measured from vertical; for the forearm, θ2 is measured from the upper arm's direction, so from vertical the forearm has turned θ1 + θ2 (45° + 45° = 90°, a horizontal forearm). Quick check: if θ2 = 0° the arm is straight, so the angle is just θ1.
7. **Deriving ΔX1 and ΔY1 (10 min)** — Walk through using sin(θ1) and cos(θ1) with L1 as the hypotenuse to solve ΔX1 = L1sin(θ1) and ΔY1 = L1cos(θ1), plugging in θ1 = 45° to get 176.777mm for each.
8. **Deriving ΔX2 and ΔY2 (5 min)** — Repeat the process with L2 as the hypotenuse, this time using the forearm angle θ1+θ2 = 90°, showing ΔX2 = L2sin(θ1+θ2) = 250mm and ΔY2 = L2cos(θ1+θ2) = 0mm.
9. **Combining the terms (5 min)** — Show the final formula (X,Y) = (ΔX1+ΔX2, ΔY1+ΔY2) and complete the worked example: (X,Y) = (426.777, 176.777).
10. **Recap & wrap-up (2 min)** — Show the "Completed Variables" slide as a clean summary of the full worked problem before dismissing.

Day 2: Extending to 3D with Base Rotation

1. **Recap (3 min)** — Quickly review the 2D derivation and final formula from Day 1.
2. **Try It Yourself #1 (7 min)** — Students solve for (X,Y) with θ1=30°, θ2=30°; reveal solution (341.506, 341.506).
3. **Try It Yourself #2 (6 min)** — Students solve with θ1=60°, θ2=0°; reveal solution (433.013, 250). Use this one to highlight how a 0° angle simplifies the math.
4. **Introducing base rotation (8 min)** — Present the "Forward Kinematics With Base Rotation" section: introduce R as the horizontal distance from the end effector to the base's center point, and show how R changes (R1 < R2) as the arm extends further from the base.
5. **θ3 and the circle (7 min)** — Show that R forms a circle around the base's center point when rotated, and that θ3 (base rotation) is known via the base actuator's encoder — reinforcing the "always known" theme from Day 1.
6. **Problem-solving strategy (7 min)** — Explain that the 3D problem splits into two 2D problems: first solving for R (and Y) along the arm's own XY plane (side view), then solving for X and Z along the XZ plane (top-down view) — the exact same method from Day 1.

7. **Worked example problem (14 min)** — Walk through the full example (θ1=45°, θ2=45°, θ3=45°) step by step: reuse the Day 1 result (X,Y)=(426.777,176.777) to get R=426.78mm, then apply Z=Rsin(θ3) and X=Rcos(θ3) to find the final 3D position (X,Y,Z) = (301.78, 176.78, 301.78).
8. **Practice Problem (7 min)** — Students solve θ1=30°, θ2=30°, θ3=60°; reveal solution (170.75, 341.51, 295.75).
9. **Practice Problem 2 (6 min)** — Students solve θ1=30°, θ2=90°, θ3=30°; reveal solution (295.75, 91.51, 170.75) — note that ΔY2 comes out negative (-125mm) because θ1+θ2 = 120° tips the forearm below horizontal, a discussion point (what does that mean physically?).
10. **Desmos Calculator demo (2 min)** — Pull up the linked 3D Desmos calculator to visually confirm one of the solved problems, giving students a spatial/interactive check on their math before closing.

Assessment:

Complete the example problems in the slideshow